Algorithm 1: ZKP-Based Reacquisition Process

Require: Reacquisition request $RQ_i = \{ID_i, \rho_i, \tau_i\}$;
Batch commitments (C_R, C_E);
Proof references (π_R, π_E);
Smart Contracts (SCs);
Blockchain Ledger $\mathcal{B}$

Ensure : Immutable record of reacquisition decision D_i on $\mathcal{B}$ for non-compliant batches.

`// Step 1: Reacquisition Request Initialization`

Receive RQ_i from stakeholder $e_r \in \{S, M, A\}$ (Smelter, Manufacturer, or Auditor)
Extract batch identifier ID_i, reason ρ_i, and request timestamp τ_i
Validate trigger: $\rho_i \in \{batch\ rejection, audit\ failure, ethical\ violation\}$

`// Step 2: Retrieval of Batch References`

Query blockchain ledger $\mathcal{B}$ for existing batch record B_i
Retrieve cryptographic commitments C_R, C_E and proof references π_R, π_E
Confirm batch B_i eligibility for reacquisition processing

`// Step 3: Proof Generation or Resubmission`

if *updated compliance evidence is required* **then**
    Generate/Refresh origin proof π'_R and compliance proof π'_E
else
    Fetch and reuse existing proof references (π_R, π_E) from IPFS/Ledger

`// Step 4: Smart Contract Submission and Verification`

Submit $(RQ_i, C_R, C_E, \pi_R, \pi_E)$ to the SC_{reacq} validation logic
$V_R \leftarrow Verify(C_R, \pi_R, R)$ `// Verify Origin Authenticity`
$V_E \leftarrow Verify(C_E, \pi_E)$ `// Verify Compliance Validity`

`// Step 5: Decision Logic`

if $(V_R = 1) \wedge (V_E = 0)$ **then**
    $D_i \leftarrow$ **Approve Reacquisition** `// Valid origin, failed compliance`
else if $(V_R = 0)$ **then**
    $D_i \leftarrow$ **Reject Reacquisition** `// Invalid or unverified origin`
else
    $D_i \leftarrow$ **Flag for Audit** `// Inconclusive or edge case`

`// Step 6: Blockchain Update and Notification`

Generate reacquisition event $E_i^{rq} = (ID_i, \rho_i, \tau_i, D_i)$
Update ledger: $\mathcal{B} \leftarrow \mathcal{B} \cup E_i^{rq}$
Broadcast decision D_i to all relevant stakeholders $e_i \in E$
if $D_i = Flag\ for\ Audit$ **then**
    Initiate secondary regulatory review process R_{audit}